AFFORDABLE RENEWABLE ENERGY TRADING

Milestone: Implementation in MySQL

Group 12

Siddiq Mohiuddin Mohammed

Ketaki Salway

857-565-8408

857-3911-895

salway.k@northeastern.edu

mohammed.si@northeastern.edu

Percentage of Effort Contributed by Student1: 50%

Percentage of Effort Contributed by Student2: 50%

Signature of Student 1:

Signature of Student 2:

Submission Date: 03/23/2025

INTRODUCTION

The **SolShare Energy Trading Database** is designed to facilitate and track energy transactions between users, manage energy profiles, process payments, and handle complaints in an organized manner. This database allows households and businesses to buy, sell, and share renewable energy efficiently while maintaining a structured record of their consumption and transactions.

For this project, we have designed a relational database using SQL and populated it with realistic random data using Python. The dataset ensures foreign key consistency, logical relationships, and meaningful data distributions across all tables.

we generated records using **Python (Faker, Random, SQLite)** and inserted them into the database.

Python Code:

```
import sqlite3

import random

import faker

import datetime

import pandas as pd


# Initialize Faker for realistic names, addresses, emails, etc.

fake = faker.Faker()

faker.Faker.seed(42)

random.seed(42)


# Generate Massachusetts-based addresses

massachusetts_cities = ["Boston", "Cambridge", "Worcester", "Springfield", "Lowell", "Quincy"]

address_list = [f"{fake.street_address()}, {random.choice(massachusetts_cities)}, MA {fake.zipcode()}"
for _ in range(100)]


# Connect to SQLite database
```

```
conn = sqlite3.connect("SolShare.db")
cursor = conn.cursor()

# === STEP 1: Create Tables ===
cursor.executescript("""
CREATE TABLE Household_User (
    UserID INTEGER PRIMARY KEY,
    Name TEXT,
    Email TEXT,
    Phone TEXT,
    Consumption_Rate FLOAT,
    Address TEXT,
    Renewable_Capacity FLOAT
);

CREATE TABLE Business_User (
    UserID INTEGER PRIMARY KEY,
    Name TEXT,
    Email TEXT,
    Phone TEXT,
    Consumption_Rate FLOAT,
    Address TEXT,
    Renewable_Capacity FLOAT
);

CREATE TABLE Energy_Profile (
    ProfileID INTEGER PRIMARY KEY,
    UserID INTEGER NOT NULL,
    Energy_Produced FLOAT,
```

```
    Energy_Consumed FLOAT,
    Surplus_Energy FLOAT,
    FOREIGN KEY (UserID) REFERENCES Household_User(UserID)
);

CREATE TABLE Region (
    RegionID INTEGER PRIMARY KEY,
    Geographical_Condition TEXT,
    SupervisorID INTEGER NOT NULL,
    Region_Name TEXT
);

CREATE TABLE Employee (
    EmployeeID INTEGER PRIMARY KEY,
    Name TEXT,
    Email TEXT,
    Phone TEXT,
    ManagerID INTEGER NOT NULL
);

CREATE TABLE Energy_Transaction (
    TransactionID INTEGER PRIMARY KEY,
    Transaction_Date DATE,
    BuyerID INTEGER NOT NULL,
    SellerID INTEGER NOT NULL,
    Energy_Amount FLOAT,
    Price FLOAT,
    FOREIGN KEY (BuyerID) REFERENCES Household_User(UserID),
    FOREIGN KEY (SellerID) REFERENCES Household_User(UserID)
```

```
);

CREATE TABLE Payment (
    PaymentID INTEGER PRIMARY KEY,
    TransactionID INTEGER NOT NULL,
    Payment_Date DATE,
    Payment_Status TEXT,
    Amount_Paid FLOAT,
    FOREIGN KEY (TransactionID) REFERENCES Energy_Transaction(TransactionID)
);

CREATE TABLE Participate (
    UserID INTEGER NOT NULL,
    TransactionID INTEGER NOT NULL,
    PRIMARY KEY (UserID, TransactionID),
    FOREIGN KEY (UserID) REFERENCES Household_User(UserID),
    FOREIGN KEY (TransactionID) REFERENCES Energy_Transaction(TransactionID)
);

CREATE TABLE Complaints (
    ComplaintID INTEGER PRIMARY KEY,
    UserID INTEGER,
    Complaint_Date DATE,
    Complaint_Description TEXT,
    EmployeeID INTEGER NOT NULL,
    Status TEXT,
    Resolution_Date DATE,
    FOREIGN KEY (UserID) REFERENCES Household_User(UserID),
    FOREIGN KEY (EmployeeID) REFERENCES Employee(EmployeeID)
```

```
);

CREATE TABLE Grid (
    ConnectionID INTEGER PRIMARY KEY,
    UserConnection1 INTEGER NOT NULL,
    UserConnection2 INTEGER,
    UserConnection3 INTEGER,
    Connection_Date DATE,
    FOREIGN KEY (UserConnection1) REFERENCES Household_User(UserID)
);

CREATE TABLE Member_of (
    UserID INTEGER NOT NULL,
    ConnectionID INTEGER NOT NULL,
    PRIMARY KEY (UserID, ConnectionID),
    FOREIGN KEY (UserID) REFERENCES Household_User(UserID),
    FOREIGN KEY (ConnectionID) REFERENCES Grid(ConnectionID)
);
""")

# === STEP 2: Data generation===

# Generate Household and Business Users
household_users = []
business_users = []
user_id = 10000

for _ in range(25):
    household_users.append((user_id, fake.name(), fake.email(), fake.phone_number(),
```

```
                    round(random.uniform(2, 10), 2), random.choice(address_list),
                    round(random.uniform(0.5, 5), 2)))
    user_id += 1

for _ in range(25):
    business_users.append((user_id, fake.company(), fake.company_email(), fake.phone_number(),
                    round(random.uniform(10, 50), 2), random.choice(address_list),
                    round(random.uniform(5, 20), 2)))
    user_id += 1

cursor.executemany("INSERT INTO Household_User VALUES (?, ?, ?, ?, ?, ?, ?)", household_users)
cursor.executemany("INSERT INTO Business_User VALUES (?, ?, ?, ?, ?, ?, ?)", business_users)

# Generate Energy Profiles
energy_profiles = [(i, i, round(random.uniform(2, 15), 2), round(random.uniform(2, 12), 2),
                round(random.uniform(-5, 10), 2)) for i in range(10000, 10050)]
cursor.executemany("INSERT INTO Energy_Profile VALUES (?, ?, ?, ?, ?)", energy_profiles)

# Generate Employees
employees = [(i, fake.name(), fake.email(), fake.phone_number(), random.randint(1000, 1005)) for i in
range(2000, 2010)]
cursor.executemany("INSERT INTO Employee VALUES (?, ?, ?, ?, ?)", employees)

# Generate Transactions
transactions = []
for i in range(5000, 5080):
    buyer = random.randint(10000, 10049)
    seller = random.randint(10000, 10049)
    while seller == buyer:
```

```
        seller = random.randint(10000, 10049)

    transactions.append((i, fake.date_this_decade(), buyer, seller,
                         round(random.uniform(1, 20), 2), round(random.uniform(50, 500), 2)))

cursor.executemany("INSERT INTO Energy_Transaction VALUES (?, ?, ?, ?, ?, ?)", transactions)

# Generate Payments
payments = [(i, i, fake.date_this_decade(), random.choice(["Completed", "Pending", "Failed"]),
            round(random.uniform(50, 500), 2)) for i in range(5000, 5080)]
cursor.executemany("INSERT INTO Payment VALUES (?, ?, ?, ?, ?)", payments)

# Generate Complaints
complaints = [(i, random.randint(10000, 10049), fake.date_this_decade(), fake.sentence(),
              random.randint(2000, 2009), random.choice(["Pending", "Resolved"]), fake.date_this_decade())
              for i in range(6000, 6020)]
cursor.executemany("INSERT INTO Complaints VALUES (?, ?, ?, ?, ?, ?, ?)", complaints)

# Generate Grid Connections & Memberships
grids = [(i, random.randint(10000, 10049), random.choice([None, random.randint(10000, 10049)]),
         random.choice([None, random.randint(10000, 10049)]), fake.date_this_decade())
         for i in range(7000, 7050)]

cursor.executemany("INSERT INTO Grid VALUES (?, ?, ?, ?, ?)", grids)

# Commit and Close
conn.commit()
conn.close()
```

OUR TABLES

Business User

UserID	Name	Email	Phone	Consumption_Rate	Address
54341	Acme Corporation	acme.corporation@example.com	617-555-0201	210.00	10 Corporate Plaza, Boston, MA, USA
54342	Beta Solutions	beta.solutions@example.com	617-555-0202	220.00	20 Business Rd, Cambridge, MA, USA
54343	Gamma Industries	gamma.industries@example.com	617-555-0203	205.50	30 Industrial Way, Worcester, MA, USA
54344	Delta Enterprises	delta.enterprises@example.com	617-555-0204	215.75	40 Commerce St, Springfield, MA, USA
54345	Epsilon Trading	epsilon.trading@example.com	617-555-0205	230.00	50 Market Ave, Lowell, MA, USA
54346	Zeta Logistics	zeta.logistics@example.com	617-555-0206	240.50	60 Innovation Dr, Lawrence, MA, USA
54347	Eta Manufacturing	eta.manufacturing@example.com	617-555-0207	225.25	70 Enterprise Blvd, Newton, MA, USA
54348	Theta Services	theta.services@example.com	617-555-0208	235.00	80 Venture St, Quincy, MA, USA
54349	Iota Retail	iota.retail@example.com	617-555-0209	210.75	90 Executive Ave, Brockton, MA, USA
54350	Kappa Technologies	kappa.technologies@example.com	617-555-0210	220.50	100 Trade St, Fall River, MA, USA

Household User

UserID	Name	Email	Phone	Consumption_Rate	Address	Renewable_Capacity
54321	Alice Johnson	alice.johnson@example.com	617-555-0101	150.75	123 Main St, Boston, MA, USA	25.50
54322	Brian Smith	brian.smith@example.com	617-555-0102	160.20	456 Elm St, Cambridge, MA, USA	30.00
54323	Carol Davis	carol.davis@example.com	617-555-0103	140.00	789 Oak St, Worcester, MA, USA	22.30
54324	David Brown	david.brown@example.com	617-555-0104	180.45	101 Pine St, Springfield, MA, USA	35.00
54325	Emily Wilson	emily.wilson@example.com	617-555-0105	200.00	202 Maple Ave, Lowell, MA, USA	28.75
54326	Frank Miller	frank.miller@example.com	617-555-0106	175.50	303 Cedar Rd, Lawrence, MA, USA	20.50
54327	Grace Taylor	grace.taylor@example.com	617-555-0107	190.00	404 Birch Ln, Newton, MA, USA	33.00
54328	Henry Anderson	henry.anderson@example.com	617-555-0108	165.25	505 Walnut St, Quincy, MA, USA	27.00
54329	Irene Thomas	irene.thomas@example.com	617-555-0109	155.75	606 Cherry Ave, Brockton, MA, USA	26.50
54330	Jack Moore	jack.moore@example.com	617-555-0110	185.00	707 Poplar Dr, Fall River, MA, USA	32.00
54331	Karen Martin	karen.martin@example.com	617-555-0111	170.50	808 Ash St, Lynn, MA, USA	21.50

Complaints

ComplaintID	UserID	Complaint_Date	Complaint_Description	EmployeeID	Status	Resolution_Date
7001	54321	2025-03-15	Billing error	1002	Resolved	2025-03-16
7002	54322	2025-03-16	Service outage	1003	Pending	NULL
7003	54323	2025-03-17	Meter malfunction	1004	Resolved	2025-03-18
7004	54324	2025-03-18	Incorrect charges	1005	Resolved	2025-03-19
7005	54325	2025-03-19	Customer service issue	1002	Pending	NULL
7006	54326	2025-03-20	Delayed response	1006	Resolved	2025-03-21
7007	54327	2025-03-21	Overbilling	1007	Pending	NULL
7008	54328	2025-03-22	Technical issue	1008	Resolved	2025-03-23
7009	54329	2025-03-23	Installation error	1009	Resolved	2025-03-24
7010	54330	2025-03-24	Payment discrepancy	1010	Pending	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Employee

EmployeeID	Name	Email	Phone	ManagerID
1001	John Manager	john.manager@example.com	617-555-1111	1001
1002	Lisa Supervisor	lisa.supervisor@example.com	617-555-2222	1001
1003	Mark Coordinator	mark.coordinator@example.com	617-555-3333	1001
1004	Nina Analyst	nina.analyst@example.com	617-555-4444	1001
1005	Oscar Operator	oscar.operator@example.com	617-555-5555	1002
1006	Paul Executive	paul.executive@example.com	617-555-6666	1002
1007	Queen Executive	queen.executive@example.com	617-555-7777	1002
1008	Rick Director	rick.director@example.com	617-555-8888	1003
1009	Susan Director	susan.director@example.com	617-555-9999	1003
1010	Tom Manager	tom.manager@example.com	617-555-1010	1001
NULL	NULL	NULL	NULL	NULL

Energy Profile

ProfileID	UserID	Energy_Produced	Energy_Consumed	Surplus_Energy
1	54321	160.00	150.75	9.25
2	54322	170.00	160.20	9.80
3	54323	150.00	140.00	10.00
4	54324	190.00	180.45	9.55
5	54325	210.00	200.00	10.00
6	54326	185.00	175.50	9.50
7	54327	200.00	190.00	10.00
8	54328	175.00	165.25	9.75
9	54329	165.00	155.75	9.25
10	54330	195.00	185.00	10.00
11	54331	180.00	170.50	9.50

Energy Transaction

Transaction_ID	Transaction_Date	BuyerID	SellerID	Energy_Amount	Price
5001	2025-03-01	54321	54341	50.00	100.00
5002	2025-03-01	54322	54342	55.00	110.00
5003	2025-03-01	54323	54343	60.00	120.00
5004	2025-03-01	54324	54344	65.00	130.00
5005	2025-03-01	54325	54345	70.00	140.00
5006	2025-03-01	54326	54346	75.00	150.00
5007	2025-03-02	54327	54347	80.00	160.00
5008	2025-03-02	54328	54348	85.00	170.00
5009	2025-03-02	54329	54349	90.00	180.00
5010	2025-03-02	54330	54350	95.00	190.00
5011	2025-03-03	54331	54351	100.00	200.00

Grid

ConnectionID	UserConnection1	UserConnection2	UserConnection3	Connection_Date
8001	54321	54322	54323	2025-03-20
8002	54324	54325	54326	2025-03-20
8003	54327	54328	54329	2025-03-20
8004	54330	54331	54332	2025-03-20
8005	54333	54334	54335	2025-03-20
8006	54336	54337	54338	2025-03-20
8007	54339	54340	54341	2025-03-20
8008	54342	54343	54344	2025-03-20
8009	54345	54346	54347	2025-03-20
8010	54348	54349	54350	2025-03-20
8011	54351	54352	54353	2025-03-20

Member of

UserID	ConnectionID
54321	8001
54322	8001
54323	8001
54324	8002
54325	8002
54326	8002
54327	8003
54328	8003
54329	8003
54330	8004
54331	8004

Participation

UserID	TransactionID
54321	5001
54321	5021
54321	5041
54321	5061
54322	5002
54322	5022
54322	5042
54322	5062
54323	5003
54323	5023
54323	5043

Payment

PaymentID	TransactionID	Payment_Date	Payment_Status	Amount_Paid
6001	5001	2025-03-02	Completed	100.00
6002	5002	2025-03-02	Completed	110.00
6003	5003	2025-03-02	Completed	120.00
6004	5004	2025-03-02	Completed	130.00
6005	5005	2025-03-02	Completed	140.00
6006	5006	2025-03-02	Completed	150.00
6007	5007	2025-03-03	Completed	160.00
6008	5008	2025-03-03	Completed	170.00
6009	5009	2025-03-03	Completed	180.00
6010	5010	2025-03-03	Completed	190.00
6011	5011	2025-03-04	Completed	200.00

Region

RegionID	Geographical_Condition	SupervisorID	Region_Name
1	Urban	1002	Eastern Boston
2	Suburban	1003	Western Cambridge
3	Rural	1004	Northern Worcester
4	Coastal	1005	Southern Salem
5	Inland	1006	Central Lowell
6	Mountain	1007	Highland Newton
NULL	NULL	NULL	NULL

SQL QUERIES FOR DATA ANALYSIS

the database allows for various analytical queries to gain insights into energy transactions and user participation. below are some sample queries:

a. **List Top 10 Users with Highest Surplus Energy**

```
SELECT ep.UserID, ep.Energy_Produced, ep.Energy_Consumed,
ep.Surplus_Energy
FROM Energy_Profile ep
ORDER BY ep.Surplus_Energy DESC
LIMIT 10;
```

UserID	Energy_Produced	Energy_Consumed	Surplus_Energy
54333	220.00	210.00	10.00
54323	150.00	140.00	10.00
54336	175.00	165.00	10.00
54325	210.00	200.00	10.00
54335	190.00	180.00	10.00
54327	200.00	190.00	10.00
54337	165.00	155.00	10.00
54332	170.00	160.00	10.00
54330	195.00	185.00	10.00
54334	205.00	195.00	10.00

b. **Average Energy Consumption by User Type**

```
-- Household Users average consumption
SELECT AVG(Consumption_Rate) AS AvgConsumption_Household
FROM Household_User;

-- Business Users average consumption
SELECT AVG(Consumption_Rate) AS AvgConsumption_Business
```

FROM Business_User;

	AvgConsumption_Business
▶	223.850000

c. Total Revenue Generated Over a Period

```
SELECT SUM(Price) AS TotalRevenue
FROM Energy_Transaction
WHERE Transaction_Date BETWEEN '2025-03-01' AND '2025-03-31';
```

	TotalRevenue
▶	16060.00

d. Count of Users in Each Grid Connection

```
SELECT m.ConnectionID, COUNT(m.UserID) AS UserCount
FROM Member_of m
GROUP BY m.ConnectionID
ORDER BY UserCount DESC;
```

	ConnectionID	UserCount
▶	8001	3
	8002	3
	8003	3
	8004	3
	8005	3
	8006	3
	8007	3
	8008	3
	8009	3
	8010	3
	8011	3

e. **List Grid Connections with Their Member Users**

```
SELECT g.ConnectionID, g.Connection_Date, m.UserID
FROM Grid g
JOIN Member_of m ON g.ConnectionID = m.ConnectionID
ORDER BY g.ConnectionID, m.UserID;
```

ConnectionID	Connection_Date	UserID
8001	2025-03-20	54321
8001	2025-03-20	54322
8001	2025-03-20	54323
8002	2025-03-20	54324
8002	2025-03-20	54325
8002	2025-03-20	54326
8003	2025-03-20	54327
8003	2025-03-20	54328
8003	2025-03-20	54329
8004	2025-03-20	54330
8004	2025-03-20	54331

f. **Customer Service and Complaint Analysis**

```
SELECT EmployeeID, COUNT(*) AS PendingComplaints
FROM Complaints
WHERE Status = 'Pending'
GROUP BY EmployeeID
ORDER BY PendingComplaints DESC;
```

EmployeeID	PendingComplaints
1002	1
1003	1
1007	1
1010	1